

Loan Approval Prediction - Short Report

Objective

Build a supervised machine learning model to predict whether a loan will be approved using borrower information.

Preprocessing

- Removed Loan_ID because it is only an identifier.
- Converted target variable Loan_Status: Y = 1, N = 0.
- Filled numeric missing values using median.
- Filled categorical missing values using most frequent value.
- Encoded categorical variables using One-Hot Encoding.
- Scaled numeric variables using StandardScaler.
- Used train-test split with stratification.

Class Imbalance Handling

The target is imbalanced, with more approved loans than rejected loans. SMOTE was applied only to the training data to avoid data leakage.

Model Comparison

Test-set results from this dataset:

Model	Precision	Recall	F1	ROC-AUC
Logistic Regression + SMOTE	0.862	0.882	0.872	0.876
Decision Tree + SMOTE	0.802	0.859	0.830	0.700
Random Forest + SMOTE	0.833	0.882	0.857	0.820

Trade-offs

Logistic Regression performed best overall and is easier to explain. Random Forest is useful when non-linear patterns matter, but it is less transparent. Decision Tree is simple, but weaker here.

Business Interpretation

The model predicts approval probability. A higher probability means the borrower looks similar to previously approved borrowers. Credit history, income, loan amount, employment, education and property area are useful approval signals.

Suggested Deployment Threshold

Start with threshold = 0.50.

Threshold comparison for the best model:

Threshold	Precision	Recall	F1
0.30	0.832	0.988	0.903
0.40	0.830	0.976	0.897

0.50	0.862	0.882	0.872
0.60	0.933	0.659	0.772
0.70	0.914	0.376	0.533

If the business wants more approvals and accepts more risk, use 0.40. If the business wants safer approvals, use 0.60. Without real default-loss cost data, 0.50 is the most practical starting threshold.